Institute for
Data Science Foundations
E-21

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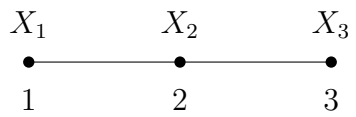
Lecture 08

In this lecture, we study exponential families that are associated with undirected graphs, so called *graphical (statistical) models*. We begin with an instructive simple example. Consider the following graph:



The node set $V = \{1, 2, 3\}$ is associated with three random variables with distribution

$$p = \sum_{x_1,x_2,x_3 \in \{0,1\}} p(x_1, x_2, x_3)\ \delta^{x_1,x_2,x_3} \in \mathcal{P}_+\left(\{0,1\}^3\right).$$

We assume that p is an element of the following exponential family which is coupled with the graph:

$$\left\{ \frac{1}{Z(\theta)} e^{\theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_{12} x_1 x_2 + \theta_{23} x_2 x_3} : \theta_1, \theta_2, \theta_3, \theta_{12}, \theta_{23} \in \mathbb{R} \right\}$$

The dimension of this exponential family is 5, and it is living in a 7-dimensional simplex. What is so special about this exponential family? We answer this question by calculating the following conditional probability:

$$\begin{aligned}
p(x_1, x_3|x_2) =& \frac{p(x_1, x_2, x_3)}{p(x_2)} = \frac{p(x_1, x_2, x_3)}{\sum_{x_1', x_3'} p(x_1', x_2, x_3')} \\
=& \frac{e^{\theta_1 x_1 + \cancel{\theta_2 x_2} + \theta_3 x_3 + \theta_{12} x_1 x_2 + \theta_{23} x_2 x_3}}{\sum_{x_1', x_3'} e^{\theta_1 x_1' + \cancel{\theta_2 x_2} + \theta_3 x_3' + \theta_{12} x_1' x_2 + \theta_{23} x_2 x_3'}} \\
=& \frac{e^{\theta_1 x_1 + \theta_{12} x_1 x_2} \cdot e^{\theta_3 x_3 + \theta_{23} x_2 x_3}}{\sum_{x_1', x_3'} e^{\theta_1 x_1' + \theta_{12} x_1' x_2} \cdot e^{\theta_3 x_3' + \theta_{23} x_2 x_3'}} \\
=& \frac{e^{\theta_1 x_1 + \theta_{12} x_1 x_2}}{\sum_{x_1'} e^{\theta_1 x_1' + \theta_{12} x_1' x_2}} \cdot \frac{e^{\theta_3 x_3 + \theta_{23} x_2 x_3}}{\sum_{x_3'} e^{\theta_3 x_3' + \theta_{23} x_2 x_3'}} \\
=& \Phi_{12}(x_1, x_2) \cdot \Phi(x_2, x_3)
\end{aligned}$$

where the Φs are functions that only depend on a subset of arguments that are connected by edges. One can prove

$$\begin{aligned}
\Phi_{12}(x_1, x_2) &= p(x_1|x_2) \\
\Phi_{23}(x_2, x_3) &= p(x_3|x_2) \qquad \text{(Exercise)}
\end{aligned}$$

Thus, X_1 and X_3 are conditionally independent given X_2. In symbols

$$X_1 \perp\!\!\!\perp X_3 | X_2. \tag{1}$$

As we will see, this is due to the fact that we cannot go from 1 to 3 or from 3 to 1 without going through 2. The graphical structure implies a number of conditional independence statements similar to (1). These statements give rise to an efficient sampling method associated with the graph.

We now begin with the general theory of graphical models. Consider an undirected graph $G = (V, E)$ with *node set* V and *edge set* $E \subseteq \binom{V}{2}$. (Here, $\binom{V}{2}$ denotes the set of subsets of V with cardinality 2). If $\{v, w\} \in E$ then we write $v \sim w$ and v and w are called *neighbours*. Given a node v, the set $\{w \in V : v \sim w\}$ of neighbours is called the boundary of v and denoted by $\mathrm{bd}(v)$. For an arbitrary subset A of V, we define the boundary

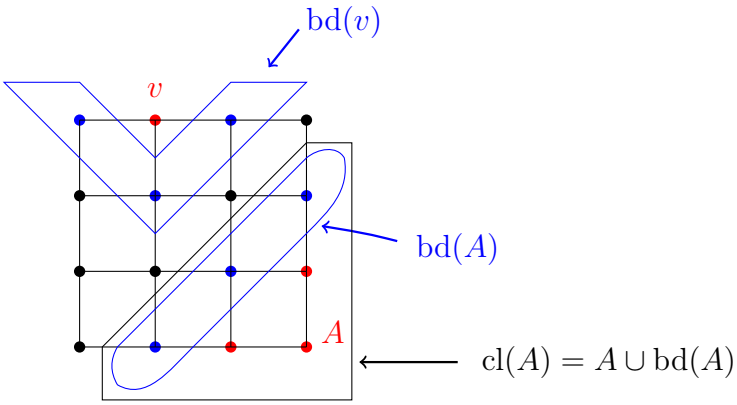
$$\mathrm{bd}(A) := \bigcup_{v \in A} \big(\mathrm{bd}(v) \setminus A\big)$$

of A and its closure

$$\mathrm{cl}(A) := A \cup \mathrm{bd}(A).$$

Note that $A \cap \mathrm{bd}(A) = \emptyset$

Example.



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A *path* in V is a sequence $\gamma = (v_1, ..., v_n)$ satisfying $v_i \sim v_{i+1}$ for all $i = 1, ..., n-1$. Let A, B, and S be three disjoint subsets of V. We say that S separates A from B if for every path $\gamma = (v_1, ..., v_n)$ with $v_1 \in A$ and $v_n \in B$ there is $v_i \in S$. Note that $\mathrm{bd}(A)$ separates A from $V \setminus \mathrm{cl}(A)$. (Exercise)

A subset C of V is called a *clique* of G, if for all $v, w \in C$, $v \neq w$, it holds that $v \sim w$.

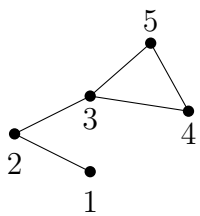
Definition. Let G be a graph, and let $\mathcal{C}(G)$ be the set of cliques of G. Then the exponential family consisting of probability measures

$$p = \frac{1}{Z} e^{\sum_{c \in \mathcal{C}(G)} f_c(x_c)}$$

is called a *graphical model*. Here, f_c depends only in the states $x_i, i \in C$. ◊

Example.

G :



Cliques:

$\{1\}, \{2\}, \{3\}, \{4\}, \{5\},$

$\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 5\}, \{3, 5\},$

$\{3, 4, 5\}$

$$\frac{1}{Z}e^{f_1(x_1)+f_2(x_2)+f_3(x_3)+f_4(x_4)+f_5(x_5)+f_{\{1,2\}}(x_{1,2})+f_{\{2,3\}}(x_{2,3})+f_{\{4,5\}}(x_{4,5})+f_{\{3,5\}}(x_{3,5})+f_{\{3,4,5\}}(x_{3,4,5})}$$

If we go through all possible functions, $f_1, f_2, f_3, f_4, f_5, f_{\{1,2\}}, f_{\{2,3\}}, f_{\{3,4\}}, f_{\{4,5\}}, f_{\{3,5\}}, f_{\{3,4,5\}}$, this forms an exponential family of dimension 11, living in a simplex of dimension 31.

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A graph encodes natural conditional independence properties, so-called *Markov properties*, which are satisfied by all probability measures of the corresponding graphical model.

$$X_A \perp\!\!\!\perp X_B \mid X_C,$$

in words, "X_A and X_B are (stochastically) independent given X_C," if

$$p(x_A, x_B|x_c) = p(x_A|x_c) \cdot p(x_B|x_C) \quad \text{for } p(x_C) > 0.$$

Definition. Let G be a graph with node set V, and let p be a distribution of $X_i, i \in V$. Then we say that p satisfies the
(G) *global Markov property*, if

$$A, B, S \subseteq V \text{ disjoint}, S \text{ separates } A \text{ from } B \quad \Rightarrow \quad X_A \perp\!\!\!\perp X_B|X_S;$$

(S) *setwise Markov property*, if

$$A \subseteq V \quad \Rightarrow \quad X_A \perp\!\!\!\perp X_{V\setminus \mathrm{cl}(A)}|X_{\mathrm{bd}(A)};$$

(L) *local Markov property*, if

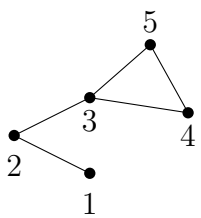
$$v \in V \quad \Rightarrow \quad X_v \perp\!\!\!\perp X_{V\setminus \mathrm{cl}(v)}|X_{\mathrm{bd}(v)};$$

(P) *pairwise Markov property*, if

$$v, w \in V, v \nsim w, \quad \Rightarrow \quad X_v \perp\!\!\!\perp X_w|X_{V\setminus\{v,w\}};$$

◊

Example.



(G) $X_5 \perp\!\!\!\perp X_{\{1,2\}}|X_3$

(S) $X_{\{3,4\}} \perp\!\!\!\perp X_1|X_{\{2,5\}}$

(L) $X_2 \perp\!\!\!\perp X_{\{4,5\}}|X_{\{1,3\}}$

(P) $X_2 \perp\!\!\!\perp X_4|X_{\{1,3,5\}}$

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Proposition.

$$(G) \quad \Rightarrow \quad (S) \quad \Rightarrow \quad (L) \quad \Rightarrow \quad (P).$$

□

We now provide a criterion for a distribution p that is sufficient for the global Markov property, and therefore also for the local and pairwise Markov property. We say that p *factorises*

according to G or satisfies the
(F) *factorisation property*, if there exist functions Φ_C, C a clique, such that

$$p(x) = \prod_{C \text{ clique}} \Phi_C(x_C).$$

Here, Φ_C only depend on $x_i, i \in C$.

Proposition.

$$(F) \quad \Rightarrow \quad (G).$$

□

Theorem. *(Hammersley–Clifford theorem) For strictly positive* p*, we have*

$$(P) \quad \Rightarrow \quad (F).$$

Thus, for a graphical model all conditions are equivalent.